Computer Engineering 4DK4
Lab 3
ALOHA Medium Access Control and Packet Reservation

Note: Lab 3 and Lab 4 (Selective-Repeat ARQ protocol) are not directly related, but Lab 5 integrates these two protocols to support an application-layer protocol. If you plan to use Python in the later labs, you should consider using SimPy for Lab 3. However, no SimPy implementation of ALOHA will be provided. For Lab 4, neither C nor SimPy code will be provided.

This lab investigates the performance of the ALOHA media access control protocol. In the provided ALOHA simulation code, the random backoff used by the stations is exponentially distributed with a fixed mean. Performance results are obtained using this simulation for various combinations of numbers of stations and mean backoff. Later in the lab, this is changed so that the mean backoff is chosen dynamically for a given packet using a binary exponential backoff algorithm. The code is then modified so that one station functions differently than the others, which results in preferential treatment. A Slotted-ALOHA version of the protocol is then designed and its performance is characterized. Finally, a packet reservation protocol is simulated where reservation mini-slots are accessed using the S-ALOHA protocol.

1 Preparation

1. An electronic copy of the simulation program must first be obtained. The program consists of a number of C code and header files. You must also include the `simlib.h`, `simlib.c` and `trace.h` Simlib library files when you compile and link the simulation. A zip file of everything you need is available on the course web site.

2. As in the other labs first familiarize yourself with the code and with running the simulation. Make sure that you understand how this simulation works. Set `MEAN_PACKET_DURATION` to 1. *In all the experiments make sure you include runs using your McMaster student ID number as the random number generator seed.*

2 Experiments

1. Set the `NUMBER_OF_STATIONS` to 10, and generate mean delay (D) versus arrival rate (λ) curves for `MEAN_BACKOFF_DURATION` values of 5, 10, and 20. Repeat these experiments when the `NUMBER_OF_STATIONS` is set to 5 using `MEAN_BACKOFF_DURATION` values of 3, 5 and 7.

2. Modify the code so that you can also generate throughput (S) versus arrival rate (λ) curves. Set `NUMBER_OF_STATIONS` to 10 and generate the throughput (S) versus arrival rate (λ) curves for `MEAN_BACKOFF_DURATION` values of 5, 10, and 20. Repeat these experiments when the `NUMBER_OF_STATIONS` is set to 5 using `MEAN_BACKOFF_DURATION` values of 3, 5 and 7.

   Generate i) the mean delay (D) versus throughput (S) curves and ii) the throughput (S) versus aggregate traffic load (G) curves for the above parameter settings.

   Compare your throughput (S) versus aggregate traffic load (G) curves with the curve in the lecture notes, describe their similarities and differences, and explain the possible reasons for any differences you observe.

3. Modify the simulation so that the stations use a *binary exponential backoff*, such as that used in Ethernet. The backoff is chosen uniformly in the range $[0, 2^{N_c})$ packet transmission times, where N_c is the number of collisions that the packet has suffered. Note that in the simulation code there is a packet struct member that already keeps track of the number of collisions that a packet has had. Generate a mean delay (D) versus throughput (S) curve for 5 and 10 stations. Compare your results with the D-versus-S curves obtained using fixed mean backoffs in Part 2.

4. Modify the simulation from Part 3 so that one particular station always re-transmits an unsuccessful packet in the very next slot (i.e., it persists and does not backoff at all). Compare the mean delay and throughput of this station with those of the other stations.

5. Change the simulation so that it operates using Slotted-ALOHA. The best way to do this is to implement a guard time where transmissions start and end ϵ seconds after and before the actual time slot boundaries as shown in Figure 1. Compare the results you get with S-ALOHA to those generated in Part 3. In your results use values of ϵ that are small enough that they have only a negligible affect on the overhead.

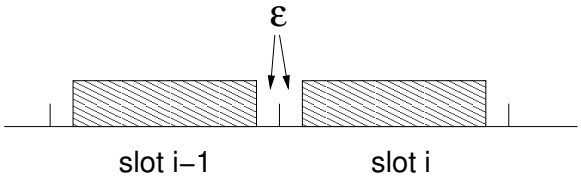


Figure 1: Slotted ALOHA with Guard Times

6. Consider a system where `NUMBER_OF_STATIONS` stations are accessing a shared data channel. When a station has a packet to send (i.e., the packet is at the front of the station's FIFO buffer), it must first transmit successfully on a separate slotted-ALOHA reservation channel. The S-ALOHA channel consists of continuous reservation mini-slots of duration X_r seconds. Once the S-ALOHA transmission is successful, the packet is placed in FCFS order for transmission on the data channel.

   Generate a simulation for this situation. Assume that packets arrive according to a Poisson process as in the supplied code. Also assume that data packet lengths are exponentially distributed, with a mean of X seconds. Generate performance results that show the mean delay

experienced by the data packets as a function of other system parameters. Also, generate results that show the capacity of the system for several different values of the ratio X/X_r.

Explain the results that you obtain in all the experiments.

3 Writeup

Each group (maximum of three students) should submit a single writeup for the lab. The writeup should clearly document your experiments, results, and observations.

1. Experimental setup: Clearly describe the simulation setup and parameter values used for each experiment, including the random number generator seeds used to generate the reported results. Describe any assumptions or design choices you made when modifying the simulation.

2. Results: Include all plots requested in Section 2. Unless otherwise specified, select appropriate parameter ranges so that the important operating regions and performance trends of the systems can be clearly observed. Include sufficient simulation data points to support the conclusions drawn from the results.

3. Analysis and discussion: For each experiment, describe the important observations from your results and explain the reasons for the observed behavior. Where comparisons are requested, clearly identify and explain the similarities and differences between the systems being compared.

4. Simulation accuracy: Use a sufficiently long simulation run and, where appropriate, multiple random number generator seeds to obtain reliable results. If the results exhibit significant variability, discuss its effect on your conclusions.

5. Code modifications: Include the portions of the supplied code that you modified or added. You do not need to include unchanged portions of the supplied code.

6. Plots: Clearly label all axes, units, curves, and relevant parameter values. Use markers to identify simulation data points and appropriate fitted or theoretical curves where useful. Do not use piecewise-linear interpolation or overfit the simulation data to produce smooth curves.

The lab report is marked according to the above writeup requirements. The CEAB indicators below are provided for reference and show how selected components of the lab contribute to graduate attribute assessment.

Table 1: Lab 3 CEAB Assessment Rubric

Indicator	3 – Exceeds Expectations	2 – Meets Expectations	1 – Partially Meets Expectations	0 – Does Not Meet Expectations
2.1 Problem Analysis	Clearly identifies relevant assumptions and design choices and correctly applies appropriate ALOHA, backoff, reservation, and discrete-event simulation principles in developing and modifying the simulation models.	States reasonable assumptions and design choices and generally applies appropriate medium-access and simulation principles correctly.	Some assumptions or design choices are unclear, incomplete, or weakly connected to the engineering principles used in the simulation.	Important assumptions or design choices are missing or unreasonable, and/or the simulation is based on incorrect engineering principles.
2.2 Problem Analysis	Explains observed simulation behaviour using well-substantiated reasoning. Comparisons among protocols and operating conditions are clear and insightful, and relevant limitations of the results are recognized.	Provides reasonable explanations for the main observed trends and correctly discusses the important similarities and differences among protocols and operating conditions.	Provides mostly descriptive observations with limited supporting reasoning; important comparisons or limitations are only partially addressed.	Provides little or no meaningful analysis of the results, or explanations are inconsistent with the simulation results or underlying protocol behaviour.
3.1 Investigation	Selects appropriate arrival-rate ranges, traffic loads, backoff parameters, reservation parameters, simulation run lengths, random seeds, and numbers of data points so that important operating regions and performance trends are clearly demonstrated; variability and reliability are appropriately considered.	Uses generally appropriate parameter ranges, run lengths, random seeds, and sufficient data points to support the main conclusions.	Some choices of parameter ranges, run lengths, random seeds, or numbers of data points are insufficient, making some trends or conclusions unclear or unreliable.	Simulation settings or data-collection methods are inappropriate or insufficient to support the reported conclusions.
4.4 Design	Clearly justifies simulation design and implementation decisions, including modifications for throughput measurement, binary exponential backoff, persistent retransmission, Slotted ALOHA, and packet reservation, with thoughtful consideration of relevant assumptions, constraints, and limitations.	Provides reasonable justification for the major simulation design and implementation decisions, with appropriate consideration of assumptions and constraints.	Design and implementation decisions are described but only weakly justified; relevant assumptions, constraints, or limitations are incompletely considered.	Design or implementation decisions are unexplained, inappropriate, or inconsistent with the required system model.